Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)

by

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Candidate's Declaration

This is to certify that the work presented in this thesis entitled, **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"**, is the outcome of the research carried out by the authors under the supervision of **Md Johir Bin Alam, Professor, Department of CEE, Shahjalal University of Science & Technology, Bangladesh**, and the co-supervision of **Mohiuddin Arif, Lecturer, Department of Civil Engineering, Sylhet Engineering College, Bangladesh**.

It is also declared that neither this thesis nor any part thereof has been submitted anywhere else for the award of any degree, diploma, or other qualifications.

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Recommendation Letter from Thesis Supervisor

The thesis entitled **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"** submitted by the students:

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is a record of research work carried out under our supervision and we, hereby, approve that the report is submitted in partial fulfilment of the requirement for the award of their Bachelor's Degree.

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Signature of the Supervisor	Signature of the Co-Supervisor
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Certificate of Acceptance

The thesis titled **"Comparative Analysis of Locally Produced Bitumen and Imported Low-Cost Bitumen for Bangladesh Road Construction (Focusing on Aggregate Shape)"** submitted by Md. Rakibul Hasan Mridha and Md Sagor Hossain Saddas; Student ID: 2020333545 and 2020333553, to the Department of Civil Engineering, Sylhet Engineering College, has been accepted as satisfactory in partial fulfillment of the requirement for the Degree of Bachelor of Science in Civil Engineering and approved as to its style and contents.

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The Authors

Abstract

Many flexible pavements in Bangladesh deteriorate prematurely because of inconsistent imported bitumen and poor aggregate shape. In this study, we examined how bitumen source quality and aggregate particle shape together affect the Marshall mix properties of bituminous mixtures. We tested four mix combinations in the laboratory: locally produced bitumen and imported bitumen, each combined with either natural aggregates or a 20% cubical aggregate replacement. All specimens were compacted at 75 blows per face following Roads and Highways Department specifications for heavy traffic, and the optimum binder content was fixed at 5.0%. The results showed that local bitumen paired with 20% cubical aggregates performed best. This particular mix reached a Marshall stability of 13.3 kN (2991 lbs) with 5.20% air voids. Compared to the imported bitumen control mix (which had the lowest stability and the highest flow), stability improved by 167% and permanent deformation dropped by 50%. On the other hand, when imported bitumen was combined with cubical aggregates, air voids fell to 2.59%, which raises the risk of bleeding in the pavement surface. Overall, the data suggest that using local bitumen with shape-sorted aggregates can improve load-bearing capacity and rutting resistance in flexible pavements under subtropical conditions.

Keywords: bitumen, aggregate shape, Marshall stability, cubical aggregate, Bangladesh, road construction, flexible pavement

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1 Introduction

1.1 Background

Bangladesh has a road network of approximately 375,000 km, and more than 90% of it consists of flexible pavements built with bituminous materials. How well these pavements hold up depends largely on the quality of two components: the bituminous binder and the mineral aggregates [9]. Bitumen acts as the glue that holds aggregate particles together in the pavement. Eastern Refinery Limited (ERL) in Chittagong is the only domestic producer, and it supplies penetration-grade 60/70 bitumen at a controlled price of roughly Tk 6,300 per unit. However, domestic production does not always keep up with demand, so contractors often turn to imported bitumen from international markets.

The Roads and Highways Department (RHD) has, on several occasions, restricted certain imported bitumen grades because of quality concerns and the pavement damage they can cause [2]. Despite these restrictions, imported low-cost bitumen is still widely used in road projects across the country, mostly for short-term cost savings.

Aggregates make up about 92% to 96% of hot mix asphalt (HMA) by weight and play a major role in pavement performance [4]. Among the various aggregate properties, particle shape is particularly important because it affects how well particles interlock, how the mix compacts, and how it resists permanent deformation [7]. Cubical aggregates, which have roughly equal dimensions along all three axes, provide better mechanical interlock and internal friction than flaky, elongated, or blade-shaped particles [6]. The effect of aggregate shape on mix performance has been studied before, but very few researchers have looked at how it interacts with the quality of local versus imported bitumen, especially in Bangladesh.

1.2 Problem Statement

There is a noticeable quality difference between the bitumen refined locally at Eastern Refinery and the imported alternatives available on the open market, and this difference has real consequences for Bangladesh's roads. Imported bitumen tends to show inconsistent penetration grades and rheological properties, and this has been linked to premature pavement problems such as rutting, stripping during the monsoon season, and fatigue cracking [3]. When these distresses appear, the structural capacity of the pavement drops and maintenance costs go up, which shortens the road's useful life.

Most of the existing research treats bitumen quality and aggregate shape as separate variables. Very few studies have combined both factors in a single experiment designed around the traffic and climate conditions found in Bangladesh [5]. The subtropical monsoon climate here is harsh: summer temperatures regularly exceed 40°C and heavy rainfall persists for months, putting bituminous pavements under both thermal and moisture stress. What has not been properly examined is how bitumen source quality and aggregate particle shape work together to affect the Marshall mix design properties of flexible pavements in this region.

1.3 Objectives of the Study

The main goal of this study is to compare the performance of locally produced bitumen from Eastern Refinery with imported low-cost bitumen for road construction in

Bangladesh, paying particular attention to how the shape of aggregate particles affects the mix. The specific objectives are:

i. To compare the physical properties (penetration, softening point, ductility, specific gravity, and flash point and fire point) of local and imported 60/70 penetration-grade bitumen.

ii. To determine the optimum binder content (OBC) using the Marshall method for the control mix with local bitumen.

iii. To evaluate the effect of cubical aggregate shape (20% replacement of coarse fraction) on the Marshall mix design properties of bituminous mixes.

iv. To assess the performance of the four mix combinations in terms of Marshall stability, flow, and volumetric properties.

v. To compare the four bituminous mixes in terms of required pavement thickness, cost effectiveness, and load bearing capacity.

vi. To provide evidence-based recommendations for material selection in Bangladesh's road construction practice.

1.4 Scope and Limitations

This study was carried out entirely in the laboratory using the Marshall mix design method. We used 60/70 penetration-grade bitumen from two sources (local Eastern Refinery and imported) along with crushed stone aggregates that conform to MoRTH Grading-II for Dense Bituminous Macadam (DBM) with a nominal maximum aggregate size (NMAS) of 19 mm. Four mix combinations were tested, using both natural (as-received) aggregates and shape-sorted aggregates where 20% of the coarse fraction was replaced with cubical particles.

The following limitations are acknowledged:

- **Aggregate Morphological Evaluation:** The shape classification of coarse aggregates was solely based on the Zingg diagram, utilizing elongation and flatness ratios. The relevant shape parameters are defined as follows:

$$\text{Elongation ratio} = d_i/d_l$$
$$\text{Flatness ratio} = d_s/d_i$$
$$\text{Shape factor} = \frac{d_s}{\sqrt{d_i \cdot d_l}}$$
$$\text{Sphericity} = \sqrt[3]{\frac{d_s \cdot d_i}{d_l^2}}$$

  where d_l, d_i, and d_s represent the longest, intermediate, and shortest dimensions of the aggregate, respectively. With these ratios, the aggregates are classified into Blade shape, Disc shape, Cubical shape, and Rod shape, as shown in Table 1.1.

Table 1.1: Aggregate shape classification using Zingg's classification

Aggregate shape	Elongation ratio	Flatness ratio
Blade	$< 2/3$	$< 2/3$
Rod	$< 2/3$	$> 2/3$
Disc	$> 2/3$	$< 2/3$
Cubical	$> 2/3$	$> 2/3$

A fuller measure would have been the Particle Index (PI), which captures the combined contributions of particle shape, angularity, and surface texture to dense packing and internal resistance. We did not determine the PI or the sphericity-based metrics in this study, and this is a limitation worth noting.

- **Material Sourcing:** All aggregates came from a single quarry in the Sylhet region, so the results may differ for aggregates with different geological origins or mineralogical compositions.
- **Binder Modification:** We only tested unmodified penetration-grade binders. Polymer-modified binders (for example, adding 5% SBS polymer) were outside the scope of this work.
- **Gradation Scope:** The study covers only the 19 mm NMAS DBM gradation. Other pavement layers such as Bituminous Concrete (BC) or Stone Matrix Asphalt (SMA) were not investigated.

1.5 Thesis Organisation

The rest of this thesis is laid out as follows. Section 2 reviews the existing literature on bitumen properties, aggregate shape effects, and Marshall mix design. Section 3 covers the experimental methodology, including the materials we used, how specimens were prepared, and the testing programme. Section 4 presents the results and discusses them in light of previous research. Section 5 gives the conclusions and recommendations for future work.

2 Literature Review

2.1 Bitumen Grading and Pavement Performance in Tropical Climates

In tropical and subtropical countries with heavy traffic volumes and wide temperature swings, the choice of bituminous binder has a direct bearing on how long a pavement lasts. Imran et al. [2] developed a pavement quality index and found that both traffic growth and temperature fluctuations speed up pavement deterioration. They also noted that poor quality control during construction leads to problems like rutting and thermal cracking. Along similar lines, Werkineh and Demissie [9] studied pavement failures on the Assosa-Mendi road network and reported that weak sub-surface materials were associated with alligator cracking and pothole formation. Both studies point to the same conclusion: without temperature-resistant bituminous binders and properly graded granular bases, highway pavements in these climates will not perform well.

2.2 Aggregate Shape and Its Influence on Mix Design

The shape, angularity, and texture of coarse aggregates all affect how a hot mix asphalt (HMA) performs in terms of stability and volumetric properties. Khairandish and Chopra [3] found that flat and elongated particles tend to break under traffic loading, which lowers shear resistance and makes the mix more prone to rutting. Teja and Krishna [7] tested dense bituminous macadam (DBM) mixes with different proportions of shaped aggregates; replacing 20% of conventional aggregates with cubical particles increased Marshall stability because the cubical particles interlock better and create more internal friction, while blade-shaped aggregates made the mix harder to compact and weaker overall. Rajkumar et al. [4] reported that having more flaky and elongated aggregates increases air voids, which can shorten the fatigue life of the pavement. Sumanth et al. [6] also showed that DBM mixes made with cubical aggregates had higher rutting resistance than mixes with other particle shapes.

Verma and Khare [8] looked at crushed rock aggregates used in Indian road construction and examined how aggregate properties affect bituminous mix performance. Following the MORT&H specifications, they tested Marshall parameters (stability, flow, VMA, VFB, and air voids) for mixes with cubical, rounded, and blade-shaped particles. Their conclusion was that flakiness hurts the quality of the bituminous mix, and cubical aggregates should be preferred for better pavement performance.

2.3 Subgrade Stabilization and Economic Road Construction

One way to bring down road construction costs is to strengthen the pavement foundation. Sood and Bansal [5] tested the use of industrial by-products like lime and fly ash for stabilizing weak subgrade soils. They found that treating the subgrade with these additives raises the California Bearing Ratio (CBR) up to a certain optimal dosage. When the subgrade is stronger, the pavement layers on top do not need to be as thick, which reduces the amount of construction material needed for the project.

2.4 Influence of Bitumen Properties on Stripping and Durability

How long bitumen lasts in the field is a central concern in pavement engineering. Bagampadde et al. [1] studied the effect of different aggregate and bitumen combinations on stripping behaviour. They found that mixes using bitumen with a lower penetration grade tended to be more durable because of their greater resilient and tensile strength. That said, other bitumen characteristics such as acid number, penetration grade, and molecular size distribution had only a small effect on moisture sensitivity.

2.5 Research Gap

Previous studies have looked at the effect of cubical aggregates [6, 7] and subgrade stabilization [5] separately, but no one has examined how aggregate shape and bitumen source quality interact in the specific conditions found in Bangladesh. In particular, nobody has systematically tested what happens when you replace 20% of the coarse aggregate with cubical particles and compare local ERL bitumen against an imported 60/70 grade binder. This study was designed to fill that gap by comparing these mix variants side by side.

It is also worth noting that Bagampadde et al. [1] reported that the inherent characteristics of bitumen have only a minor influence on the moisture sensitivity of bituminous mixes. Since all the mixes in our study used aggregates from the same quarry in Sylhet, we did not carry out separate moisture damage tests. Instead, we focused on comparing the load-bearing capacity of the two bitumen types through Marshall stability testing.

3 Methodology

3.1 General

This section describes the materials used, the experimental procedures followed, and the tests carried out. The work covered material characterisation, aggregate gradation and shape classification, mix design, specimen preparation, and performance testing. Figure 3.1 shows the overall experimental methodology.

Figure 3.1: Flowchart of the experimental methodology adopted in this study.

3.2 Materials

3.2.1 Bitumen

We used two sources of 60/70 penetration-grade bitumen in this study: (i) locally produced bitumen from Eastern Refinery Limited, Chittagong, Bangladesh, and (ii) imported low-cost bitumen bought from the open market. Both binders were tested according to the relevant ASTM standards. The tests included penetration at 25°C (ASTM D5, Standard Test Method for Penetration of Bituminous Materials), softening point by the ring-and-ball method (ASTM D36, Standard Test Method for Softening Point of Bitumen), ductility at 25°C (ASTM D113, Standard Test Method for Ductility of Asphalt Materials), kinematic viscosity (ASTM D2170, Standard Test Method for Kinematic Viscosity of Asphalts), specific gravity at 25°C (ASTM D70, Standard Test Method for Density of

Semi-Solid Asphalt Binder), and flash point by the Cleveland open cup method (ASTM D92, Standard Test Method for Flash and Fire Points). The properties of the local and imported bitumen are listed in Table 3.1 and Table 3.2.

Table 3.1: Physical properties of local bitumen (Eastern Refinery, 60/70 grade).

Property	**Test Standard**	**Specification**
Penetration at 25°C (dmm)	ASTM D5	60–70
Softening Point (°C)	ASTM D36	49–56
Ductility at 25°C (cm)	ASTM D113	≥75
Kinematic Viscosity at 135°C (cSt)	ASTM D2170	≥300
Specific Gravity at 25°C	ASTM D70	0.97–1.02
Flash Point (°C)	ASTM D92	≥230

Table 3.2: Physical properties of imported low-cost bitumen (60/70 grade).

Property	**Test Standard**	**Specification**
Penetration at 25°C (dmm)	ASTM D5	60–70
Softening Point (°C)	ASTM D36	49–56
Ductility at 25°C (cm)	ASTM D113	≥75
Kinematic Viscosity at 135°C (cSt)	ASTM D2170	≥300
Specific Gravity at 25°C	ASTM D70	0.97–1.02
Flash Point (°C)	ASTM D92	≥230

3.2.2 Aggregates

Crushed stone aggregates were collected from a local quarry in the Sylhet region. We tested the aggregates for their physical and mechanical properties following the relevant standards. The tests performed were the Los Angeles Abrasion test, the Aggregate Impact Value (AIV) test, the Aggregate Crushing Value test, specific gravity, and water absorption. The results and the specification requirements are shown in Table 3.3.

Table 3.3: Physical and mechanical properties of coarse aggregates.

Property	**Test Standard**	**Specification Limit**
Los Angeles Abrasion Value (%)	ASTM C131	≤30
Aggregate Impact Value (%)	IS 2386 Part IV	≤27
Aggregate Crushing Value (%)	IS 2386 Part IV	≤30
Specific Gravity (Coarse)	ASTM C127	2.5–2.9
Specific Gravity (Fine)	ASTM C128	2.5–2.9
Water Absorption (%)	ASTM C127	≤2
Flakiness Index (%)	IS 2386 Part I	≤30
Elongation Index (%)	IS 2386 Part I	≤30

3.2.3 Mineral Filler

Stone dust was used as the mineral filler. We obtained the filler by collecting the fraction passing the 0.075 mm (No. 200) sieve from the same quarry source. The filler gradation met the requirements given in MoRTH Section 500.

3.3 Aggregate Gradation and Blending

We designed the aggregate blend to conform to the Dense Bituminous Macadam (DBM) Grading-II specification from MoRTH Table 500-10. Sieve analysis identified four material fractions, which were blended in the following proportions to stay within the gradation envelope:

- Material A: 19 mm – 12.5 mm retained fraction (53%)
- Material B: 12.5 mm – 2.36 mm retained fraction (2%)
- Material C: 2.36 mm – 0.075 mm retained fraction (43%)
- Material D: Passing 0.075 mm, mineral filler (2%)

The combined gradation and the MoRTH specification limits are given in Table 3.4. Figure 3.2 shows the gradation curve plotted against the upper and lower specification boundaries.

Table 3.4: Combined aggregate gradation for DBM Grading-II (MoRTH Table 500-10).

IS Sieve Size (mm)	Lower Limit (%)	Upper Limit (%)	Trial Mix (%)
26.5	100	100	100
19	90	100	98
13.2	71	95	82
9.5	56	80	64
4.75	38	54	46
2.36	28	42	37
0.3	7	21	16
0.075	4	8	6

3.4 Aggregate Shape Classification

We characterised the shape of the coarse aggregates retained on the 12.5 mm sieve (Material A fraction: 19 to 12.5 mm) using the Zingg diagram classification. For each particle, three orthogonal dimensions were measured with Vernier calipers: the longest diameter (d_L), the intermediate diameter (d_I), and the shortest diameter (d_S). From these measurements, two dimensionless ratios were calculated:

$$\text{Elongation Ratio} = \frac{d_I}{d_L} \tag{1}$$

$$\text{Flatness Ratio} = \frac{d_S}{d_I} \tag{2}$$

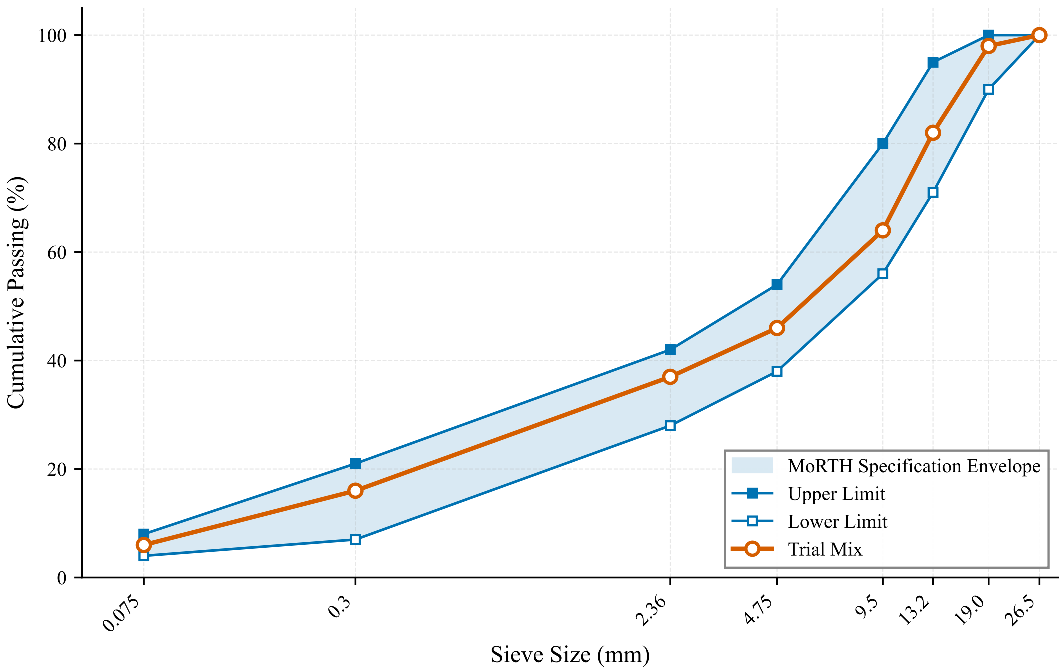

Figure 3.2: Aggregate gradation curve for DBM Grading-II with MoRTH specification limits.

Using a threshold value of 2/3 for each ratio, we classified the aggregate particles into four shape categories according to the Zingg diagram [7]: Cubical (Elongation Ratio $\geq 2/3$ and Flatness Ratio $\geq 2/3$), Rod (Elongation Ratio $< 2/3$ and Flatness Ratio $\geq 2/3$), Disk (Elongation Ratio $\geq 2/3$ and Flatness Ratio $< 2/3$), and Blade (Elongation Ratio $< 2/3$ and Flatness Ratio $< 2/3$). The Zingg classification scheme is shown in Figure 3.3.

The particles identified as cubical were separated by hand and set aside for Mix-3 and Mix-4, where they replaced 20% of the coarse aggregate (Material A) fraction by weight.

3.5 Mix Design

3.5.1 Mix Combinations

We designed four mix combinations to study how bitumen source and aggregate shape together affect the performance of bituminous mixes. Table 3.5 shows the experimental matrix.

Table 3.5: Experimental mix design matrix.

Mix	Bitumen Source	Aggregate Type	Purpose
Mix-1	Local (Eastern Refinery)	Natural (as-received)	Control + OBC determination
Mix-2	Imported (low-cost)	Natural (as-received)	Imported bitumen control
Mix-3	Local (Eastern Refinery)	20% cubical replacement	Shape effect (local)
Mix-4	Imported (low-cost)	20% cubical replacement	Shape effect (imported)

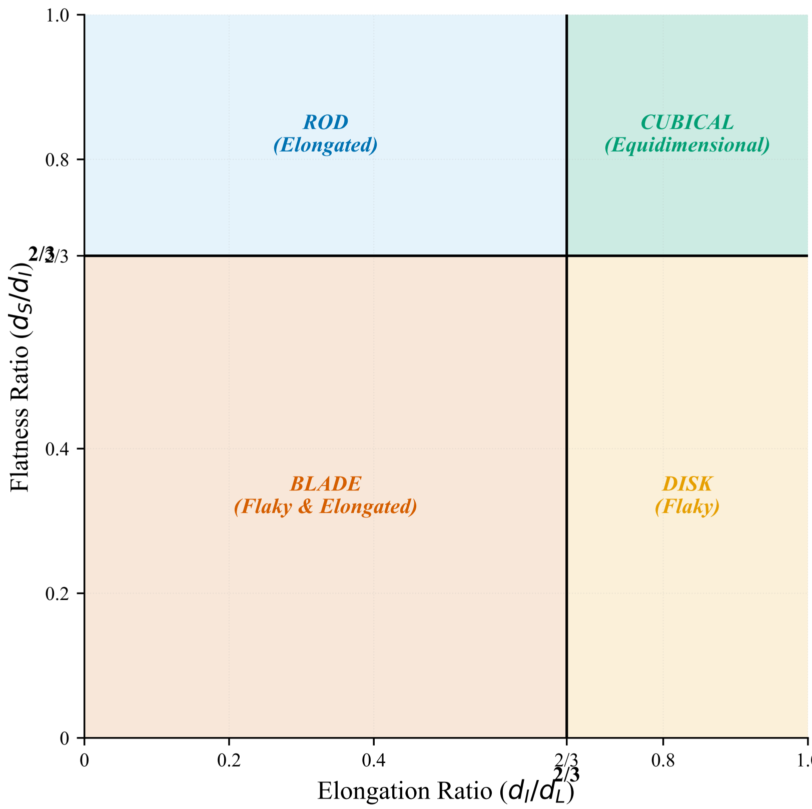

Figure 3.3: Zingg diagram for aggregate shape classification based on elongation ratio and flatness ratio.

3.5.2 Specimen Preparation

Each specimen had a total batch weight of 1200 g. Following the MoRTH specifications for Dense Bituminous Macadam (DBM) Grading-II, the material proportions were set as described in Section 3.3: 55% coarse aggregate, 43% fine aggregate, and 2% mineral filler. We heated the aggregates in an oven at 150 to 170°C before mixing. The bitumen was heated separately to 150°C so it would reach the right mixing viscosity. Once both were at the correct temperature, we combined the aggregate and the bitumen (at the optimum binder content) and mixed them thoroughly until every particle was coated.

The mix was then placed into a preheated Marshall mould (101.6 mm diameter) and rodded with a spatula to remove trapped air. Compaction was done using the Marshall hammer at 75 blows on each face, which matches the MoRTH requirement for heavy-traffic pavements [4]. After compaction, the specimens were left to cool to room temperature before being taken out of the mould.

3.5.3 Optimum Binder Content Determination

We determined the Optimum Binder Content (OBC) using Mix-1 (local bitumen with natural aggregates), which was the control mix. Five trial bitumen contents were tested: 4.0%, 4.5%, 5.0%, 5.5%, and 6.0% by weight of the total mix. For each bitumen content, three replicate specimens were made and tested for Marshall stability, flow, bulk specific gravity, and air voids.

The OBC was taken as the average of the bitumen contents corresponding to: (i) maximum Marshall stability, (ii) maximum bulk specific gravity, and (iii) 4% design air voids. The OBC turned out to be **5.0%**. We then used this value for all the remaining mixes (Mix-2, Mix-3, and Mix-4) so that the comparison would be on an equal footing.

3.6 Testing Programme

3.6.1 Marshall Stability and Flow Test

The Marshall stability and flow test followed ASTM D6927 / AASHTO T245. For each of the four mix combinations, we prepared three replicate specimens at the 5.0% OBC. Before testing, the specimens were placed in a water bath at 60°C for 30 to 40 minutes. We then recorded the Marshall stability (the maximum load the specimen could carry, in kN) and the flow value (the deformation at maximum load, in mm).

3.6.2 Volumetric Analysis

We also determined the volumetric properties of the compacted specimens for all four mixes. The following parameters were calculated:

- Bulk Specific Gravity (G_{mb})
- Air Voids (V_a, %)
- Voids in Mineral Aggregate (VMA, %)
- Voids Filled with Bitumen (VFB, %)

3.7 Summary of Test Matrix

Table 3.6 summarises the test matrix. In total, 12 Marshall specimens were prepared across the four mix combinations for the performance tests at the optimum binder content.

Table 3.6: Summary of the experimental test matrix.

Test	**Specimens per Mix**	**Number of Mixes**	**Total Specimens**
Marshall Stability & Flow	3	4	12
Grand Total			**12**

4 Results and Discussion

4.1 Properties of Aggregates

We tested the physical and mechanical properties of the coarse aggregates from the Sylhet quarry and compared them against the specification limits. Since we used a single source of crushed stone, no cross-variant comparison was needed; the results are simply checked against the maximum allowable values.

Figure 4.1 shows the results. The Aggregate Impact Value (AIV) was 18% and the Aggregate Crushing Value (ACV) was 20%, both well within the maximum limits of 27% and 30%. The Flakiness Index came out at 15% and the Elongation Index at 23%, both below the 30% maximum. This means the aggregates, even before any manual shape-sorting, already met the shape requirements.

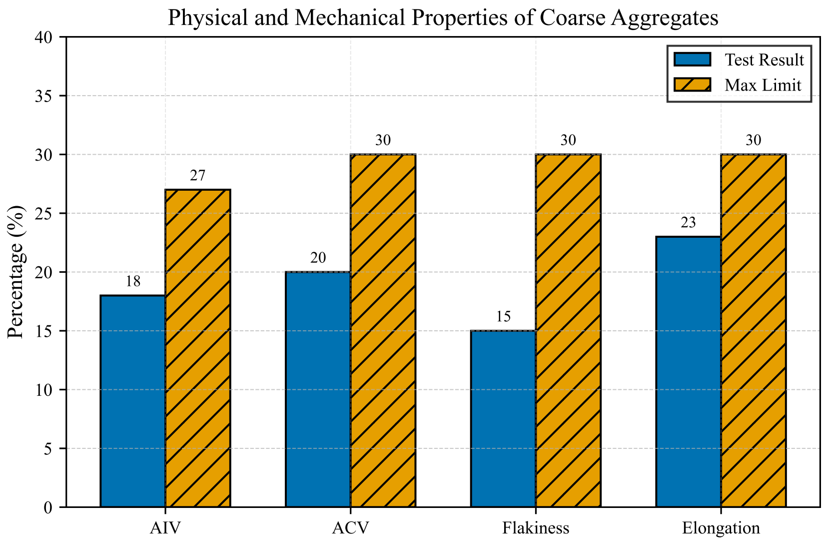

Figure 4.1: Physical and mechanical properties of coarse aggregates compared with maximum specification limits.

4.2 Properties of Bituminous Binders

Because Bangladesh's own bitumen production cannot always meet the country's demand, imported bitumen fills the gap. The trouble is that these imported binders vary in quality, and that can lead to problems like rutting and thermal cracking. To see exactly how the local and imported bitumen compare, we tested both and plotted the results.

Figure 4.2 shows the comparison. The local ERL bitumen had a higher specific gravity (1.023) than the imported bitumen (0.99). Its softening point was also higher: 52°C versus 41°C for the imported binder, which means the local bitumen resists high-temperature deformation better.

The flash point and fire point of the local bitumen (255°C and 312°C) were both higher than the imported binder's values (218°C and 253°C). Ductility was 112 cm for the local bitumen and 91 cm for the imported one. Across all the physical and rheological properties we tested, the local ERL bitumen met or exceeded the requirements.

Regional Comparative Assessment (South Asia): We also compared both binders against the highway specifications of neighbouring South Asian countries (India, Pakistan,

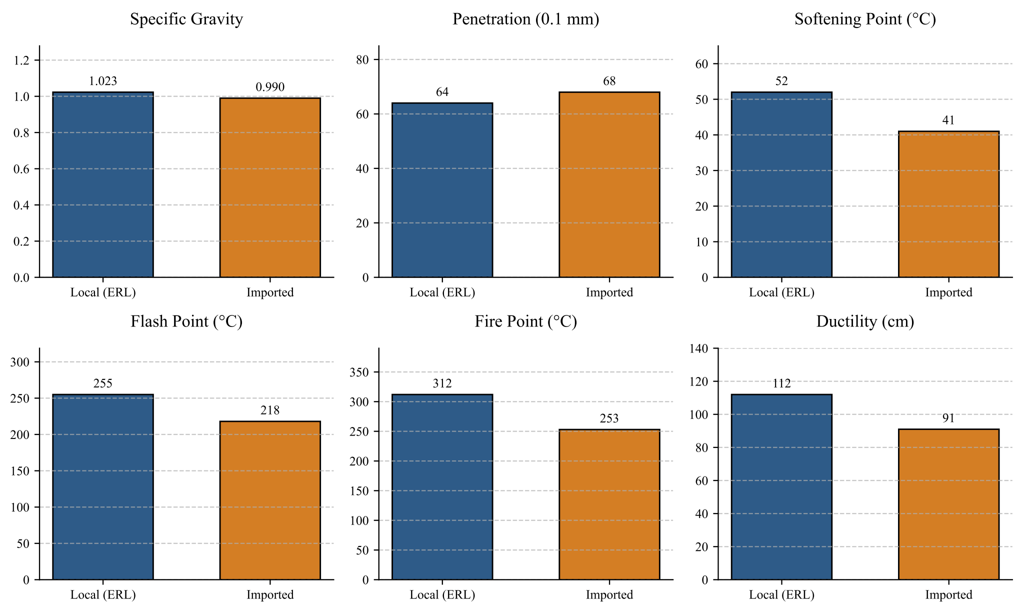

Figure 4.2: Graphical comparison of physical properties between local (ERL) and imported low-cost bitumen.

and Nepal). The details are in Table 4.1 and Figure 4.3. The local ERL bitumen, with a softening point of 52°C, passes the requirements of Pakistan's NHA (48 to 56°C), Nepal's DOR (48 to 56°C), and India's MORT&H (minimum 47°C). The imported bitumen, on the other hand, had a softening point of only 41°C, which falls below the minimum set by all of these regional authorities.

The imported bitumen's flash point (218°C) also fails to meet the 232°C minimum safety threshold used in Bangladesh, India, Nepal, and Pakistan. The local ERL bitumen's flash point of 255°C comfortably exceeds this threshold. Put simply, the imported bitumen we tested does not meet the regional specifications for use in flexible pavements.

Table 4.1: Comparative analysis of physical properties between experimental bitumen and South Asian standards.

Property	Experimental Data		National Specifications (60/70 Grade)				Standard
	Local (ERL)	Imported	RHD (BD)	MORT&H (IN)	DOR (NP)	NHA (PK)	
Penetration (dmm)	64	68	60–70	60–70	60–70	60–70	ASTM D5
Softening Point (°C)	52	**41**	48–56	47–55	48–56	48–56	ASTM D36
Ductility (cm)	112	**91**	≥100	≥75	≥100	≥100	ASTM D113
Flash Point (°C)	255	**218**	≥232	≥232	≥232	≥232	ASTM D92
Specific Gravity	1.023	**0.990**	1.01–1.06	1.01–1.06	1.01–1.06	1.01–1.06	ASTM D70

Note: Bold values indicate the imported bitumen failed to meet the specified minimum regional requirements.

4.3 Determination of Optimum Binder Content

We used the Marshall mix design method to find the Optimum Binder Content (OBC) for both the local and imported bitumen. Specimens were prepared at asphalt contents of 4.0%, 4.5%, 5.0%, 5.5%, and 6.0%. For each content, we measured the volumetric properties (air voids, VMA, VFA) and mechanical properties (Marshall stability, flow, and unit weight) to work out the best mix proportions.

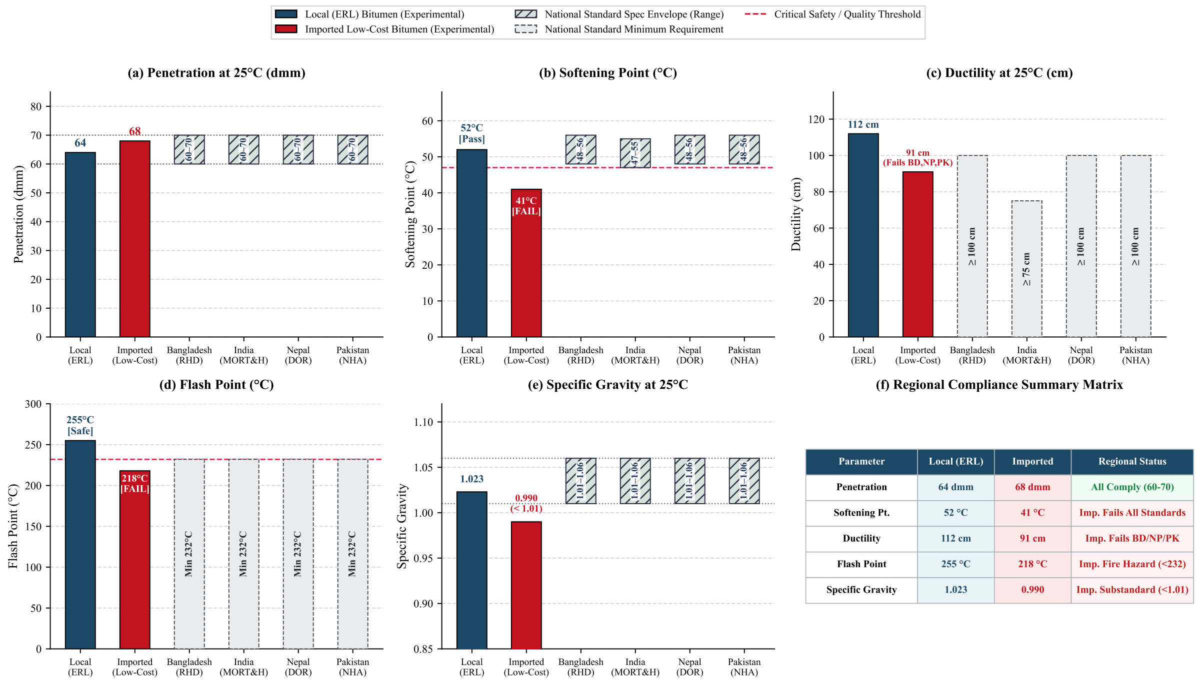

Figure 4.3: Comparative analysis of experimental physical properties of local (ERL) and imported low-cost bitumen against national highway specifications of South Asian countries (Bangladesh RHD, India MORT&H, Nepal DOR, and Pakistan NHA).

Table 4.2 and Figure 4.4 show the results. For both binders, stability peaked at a certain asphalt content and then dropped off at higher contents, while the flow value kept increasing. The target air void content (V_a) of 4.0%, which is the standard criterion for heavy-traffic pavements, corresponded to an asphalt content of about 5.0% for both the local and imported mixes. At 5.0%, the other parameters (VMA, VFA, and unit weight) all fell within the RHD specification limits. So we adopted 5.0% as the OBC for the rest of the comparative tests.

The volumetric properties of the Marshall specimens were calculated using the following equations:

$$G_{mb} = \frac{W_a}{W_{ssd} - W_w} \tag{3}$$

$$G_{mm} = \frac{P_{mm}}{\frac{P_s}{G_{se}} + \frac{P_b}{G_b}} \tag{4}$$

$$P_s = 100 - P_b \tag{5}$$

$$P_{mm} = 100 \tag{6}$$

$$G_{se} = \frac{P_{mm} - P_b}{\frac{P_{mm}}{G_{mm}} - \frac{P_b}{G_b}} \tag{7}$$

$$G_{sb} = \frac{P_1 + P_2 + P_3}{\frac{P_1}{G_1} + \frac{P_2}{G_2} + \frac{P_3}{G_3}} \tag{8}$$

$$V_a(\%) = 100 \times \frac{G_{mm} - G_{mb}}{G_{mm}} \tag{9}$$

$$\mathrm{VMA}(\%) = 100 - \frac{G_{mb} \times P_s}{G_{sb}} \tag{10}$$

$$\mathrm{VFA}(\%) = \frac{\mathrm{VMA} - V_a}{\mathrm{VMA}} \times 100 \tag{11}$$

where W_a is the weight of the specimen in air, W_{ssd} is the saturated surface dry weight, W_w is the weight in water, P_b is the percent of binder by total weight of mix, P_s is the percent of aggregate, G_b is the specific gravity of the binder, G_{se} is the effective specific gravity of the aggregate, G_{sb} is the bulk specific gravity of the aggregate blend (with P_1, P_2, P_3 and G_1, G_2, G_3 being the percentages and specific gravities of the individual aggregate fractions), G_{mm} is the theoretical maximum specific gravity of the mixture, and G_{mb} is the bulk specific gravity of the compacted mixture.

Table 4.2: Marshall mix design properties for Optimum Binder Content determination.

Bitumen Type	**AC**	W_a	W_w	W_{ssd}	G_{mb}	γ	G_{mm}	P_s	G_{sb}	V_a	**VMA**	**VFA**	H	**CF**	S_{obs}	S_{cor}	F
	4.0	1176.5	677.4	1181.2	2.330	145.72	2.520	96.0	2.684	7.54	16.66	54.75	65.0	0.932	1946	1813.67	9.5
	4.5	1216.6	704.7	1230.2	2.315	144.46	2.501	95.5	2.684	7.44	17.44	57.81	65.0	0.932	2625	2446.50	9.0
Local (ERL)	5.0	1202.3	696.9	1206.3	2.360	147.28	2.482	95.0	2.684	4.92	16.47	70.15	63.5	1.000	2500	2500.00	10.0
	5.5	1221.2	715.6	1222.3	2.410	150.38	2.464	94.5	2.684	2.19	15.15	85.54	64.3	0.950	2517	2438.60	11.0
	6.0	1230.6	722.4	1231.3	2.418	150.89	2.445	94.0	2.684	1.10	15.32	92.79	65.0	0.932	2273	2118.40	14.0
	4.0	1199.5	690.1	1204.2	2.333	145.58	2.520	96.0	2.684	7.42	16.55	55.17	64.5	0.946	1452	1373.81	10.0
	4.5	1205.9	691.5	1210.7	2.323	144.96	2.501	95.5	2.684	7.12	17.35	58.97	64.6	0.944	1133	1069.42	11.0
Imported	5.0	1206.7	701.6	1208.6	2.380	148.51	2.482	95.0	2.684	4.11	15.76	73.92	63.4	1.003	1909	1913.51	11.0
	5.5	1225.7	715.8	1226.8	2.399	149.70	2.464	94.5	2.684	2.64	15.53	83.01	63.6	0.992	2045	2028.64	12.0
	6.0	1200.0	708.1	1201.5	2.432	151.76	2.445	94.0	2.684	0.53	14.83	96.11	65.1	0.960	2274	2182.67	15.0

Note: AC = Asphalt Content (%); W_a = Specimen weight in air (g); W_w = Specimen weight in water (g); W_{ssd} = Saturated surface dry weight (g); G_{mb} = Bulk specific gravity; γ = Unit weight (lb/cft); G_{mm} = Maximum specific gravity; P_s = Percent of aggregate (%); G_{sb} = Bulk specific gravity of aggregate; V_a = Air voids (%); VMA = Voids in mineral aggregate (%); VFA = Voids filled with asphalt (%); H = Specimen height (mm); CF = Correction factor; S_{obs} = Observed stability (lbs); S_{cor} = Corrected stability (lbs); F = Flow (0.25 mm).

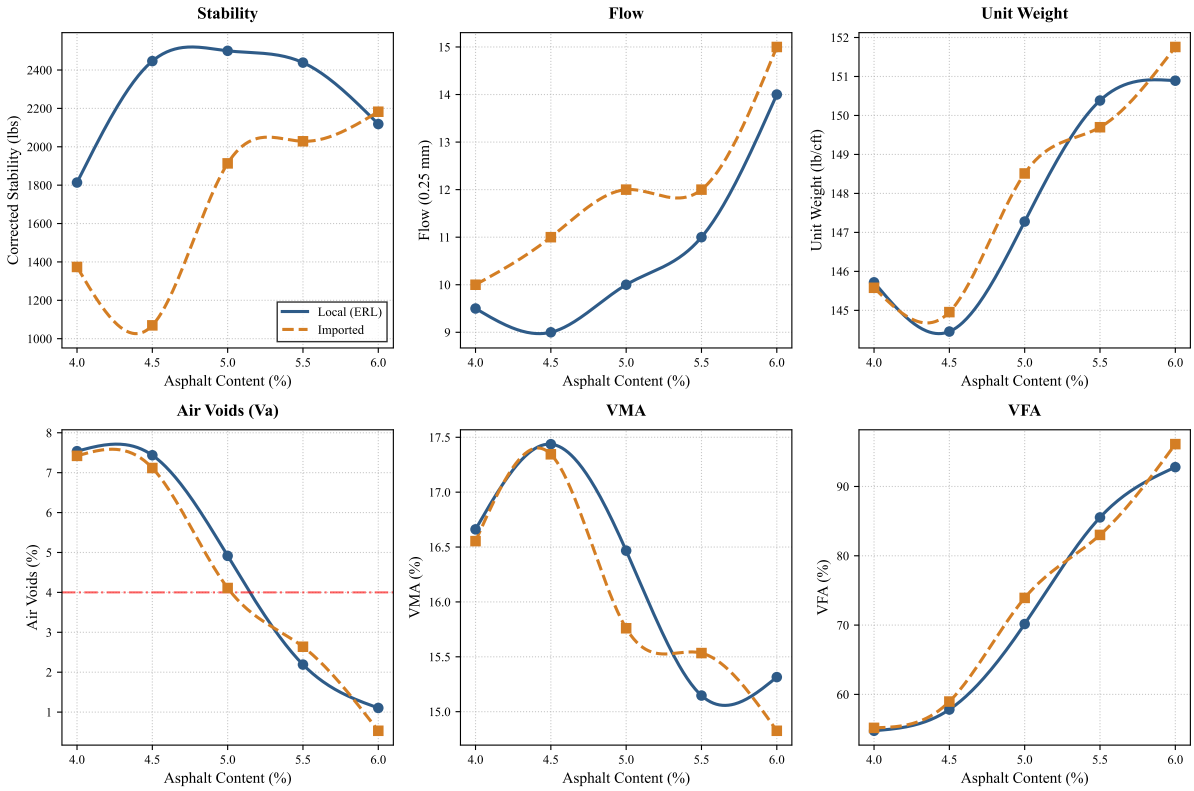

Figure 4.4: Marshall mix design properties across varying asphalt contents for Optimum Binder Content (OBC) determination.

4.4 Comparative Analysis of Bituminous Mixtures

With the OBC set at 5.0%, we then compared the four mix variants in terms of load-bearing capacity, deformation (flow), required pavement thickness, and cost-effectiveness. The Marshall properties of all four variants are summarised in Table 4.3, and Figure 4.6 plots the interaction between binder type (local vs. imported) and aggregate shape (normal vs. 20% cubical).

Table 4.3: Comparative Marshall properties of the four bituminous mix variants evaluated at 5.0% OBC.

Variant	OBC	W_a	W_w	W_{ssd}	G_{mb}	γ	G_{mm}	P_s	G_{sb}	V_a	VMA	VFA	H	CF	S_{obs}	S_{cor}	F
Local + Normal	5.0	1168.9	688.5	1176.0	2.398	149.64	2.479	95.0	2.680	3.27	15.00	78.20	61.9	1.040	1883	1958.81	10.0
Imp + Normal	5.0	1176.8	679.7	1181.0	2.348	146.52	2.469	95.0	2.680	4.90	16.77	70.78	64.6	0.994	1126	1119.40	17.0
Local + 20% Cubical	5.0	1180.2	684.3	1186.0	2.350	146.64	2.479	95.0	2.680	5.20	16.70	68.86	60.3	1.090	2744	2990.55	8.5
Imp + 20% Cubical	5.0	1250.3	733.2	1253.0	2.405	150.07	2.469	95.0	2.680	2.59	14.93	82.65	65.1	0.960	2156	2070.13	12.0

Note: See Table 4.2 for column abbreviations. OBC = Optimum Binder Content (%).

A. Overall Performance Assessment:

We looked at all four mixes across both structural and volumetric parameters. Figure 4.5 shows Marshall stability, flow value, unit weight, air voids (V_a), voids in mineral aggregate (VMA), and voids filled with asphalt (VFA) for each mix. The data in Table 4.3 show that the binder source and the aggregate shape interact in ways that matter.

Looking at the volumetric side first: how the aggregate particles arrange themselves affects how well the mix compacts. The two control mixes with normal aggregates (Mix-1 and Mix-2) had air voids within standard limits (3.27% and 4.90%) and enough VMA to be durable. But when we added 20% cubical aggregates, the volumetric response depended on which bitumen was used. With imported bitumen and cubical aggregates (Mix-4), the mix became very dense, with air voids dropping to just 2.59% and VFA reaching 82.65%. In a subtropical monsoon climate, air voids that low raise the risk of binder bleeding and plastic flow when the pavement is loaded by traffic [2].

The local ERL bitumen combined with 20% cubical aggregates (Mix-3) told a different story. It had a VMA of 16.70%, which leaves enough room for the binder film, while keeping air voids at 5.20%. As Figure 4.5 shows, this volumetric balance went hand-in-hand with higher shear resistance: the Marshall stability of this mix was 2990.55 lbs. The unit weight data also point to better densification and stronger aggregate interlock with the cubical particles. What Mix-3 shows is that it is possible to boost structural capacity without sacrificing the volumetric proportions needed for long-term fatigue life and rutting resistance.

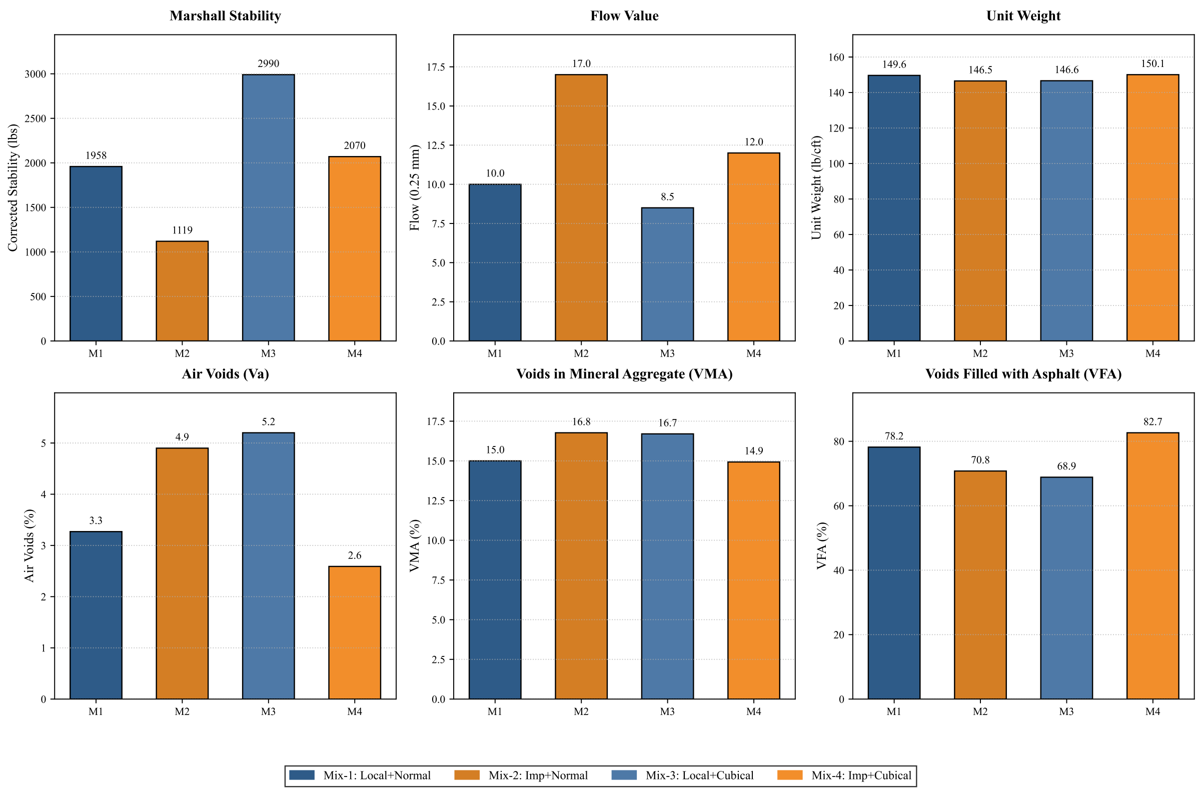

Figure 4.5: Comprehensive performance assessment of the four bituminous mix variants across key Marshall stability, flow, and volumetric parameters.

B. Load Bearing Capacity (Marshall Stability): The left subplot of Figure 4.6 shows that adding 20% cubical aggregates raised the load-bearing capacity for both binder types. The mix with local ERL bitumen and 20% cubical aggregates gave the highest Marshall stability (2991 lbs). The ERL bitumen has a lower penetration value, which makes the mix stiffer. Bagampadde et al. (2006) [1] reported something similar: bitumen with lower penetration grades tends to produce stronger mixes. At the other end, the imported bitumen with normal aggregates gave the lowest stability of all four mixes (1119 lbs).

C. Deformation (Flow Value): The right subplot of Figure 4.6 shows the flow values, measured in units of 0.25 mm. The imported bitumen with normal aggregates had the highest flow (17 units), which means this mix is the most prone to permanent deformation and rutting. The local bitumen with 20% cubical aggregates had a flow of just 8.5 units, well within the RHD limits for heavy-duty pavements.

D. Required Pavement Thickness: Under the AASHTO structural design guidelines for flexible pavements, the structural layer coefficient (a_1) of the bituminous surfacing depends on its Marshall stability and resilient modulus. Since the local bitumen with 20% cubical aggregates mix had the highest stability, it has the highest a_1 value. In practical terms, this means that to reach a given target Structural Number (SN) for a particular traffic load, this mix would need a thinner asphalt layer than any of the other three variants. The imported bitumen with normal aggregates, having the lowest structural coefficient, would need the thickest layer.

E. Cost Effectiveness: Pavement thickness has a direct effect on construction costs. Because the local bitumen and 20% cubical aggregates mix can reach the needed structural capacity with a thinner layer, it cuts down the total volume of bituminous material needed per kilometre of road. The savings on materials can offset whatever extra costs come from local refining or sorting aggregates by shape, making this mix a cost-effective option over the pavement's service life.

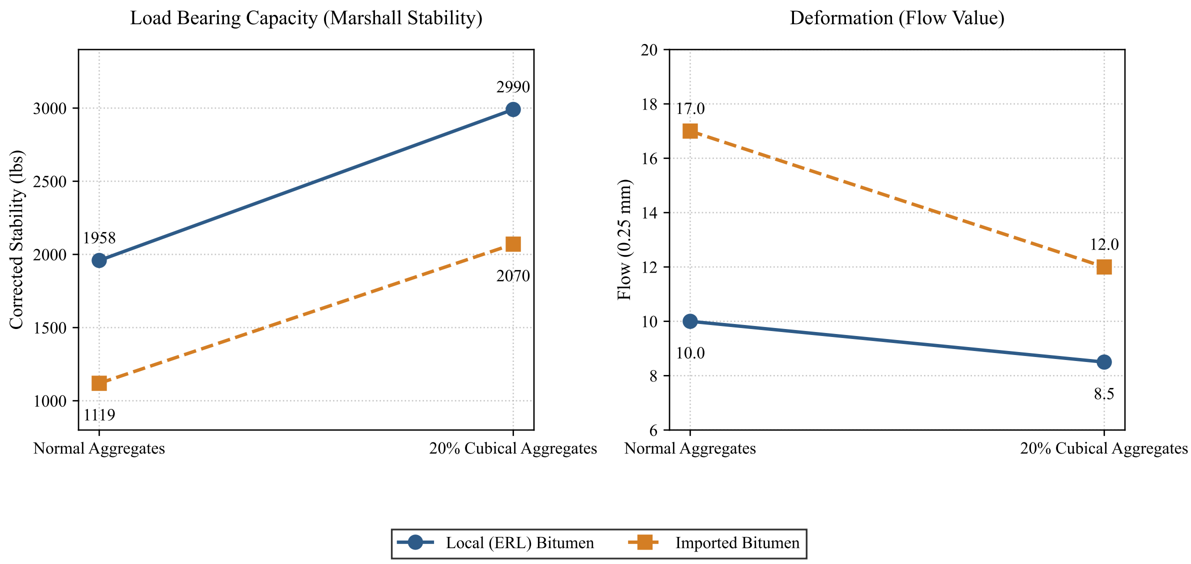

Figure 4.6: Interaction line plots comparing Marshall stability and flow across four bituminous mix variants.

4.5 Proposed Roadmap for Cost-Effective Road Construction in Bangladesh

Road construction in Bangladesh is expensive partly because of weak subgrade soils, heavy monsoon rains, and dependence on imported materials. Drawing on our findings and the existing literature, we propose a two-level strategy that could improve pavement durability while keeping costs manageable.

The idea is to combine subgrade improvement with a better surface mix:

1. **Foundation Level (Subgrade Stabilization):** The subgrade carries the load of the entire pavement structure. In Bangladesh, the subgrade is often weak, which means the base and sub-base layers have to be built thicker to compensate. Sood

and Bansal [5] showed that treating weak subgrade soils with lime and fly ash raises the CBR. A stronger subgrade means the layers above it can be thinner.

2. **Surface Level (Optimized Bituminous Mix):** On top of the stabilized foundation, the surface course needs to resist rutting and moisture damage. Our experimental results showed that combining local ERL bitumen with 20% cubical aggregates gave a Marshall stability of 2991 lbs and a flow of 8.5 units. In terms of pavement design (for example, using the AASHTO 1993 method), this high stability translates to a higher structural layer coefficient (a_1).

Techno-Economic Summary: If the foundation is strengthened with lime and fly ash and the surface course uses ERL bitumen with 20% cubical aggregates, the required structural number (SN) can be reached with thinner pavement layers. Using less aggregate and less bitumen per kilometre of road is a practical way to lower construction costs in Bangladesh without compromising the structural performance of the pavement.

5 Conclusion

5.1 Summary of Findings

This study compared locally produced Eastern Refinery bitumen with imported bitumen in a laboratory setting, looking specifically at how aggregate particle shape (20% cubical replacement) affects the Marshall mix design properties. From the experiments and volumetric analyses, the main findings are:

- The locally refined ERL bitumen had better rheological properties than the imported binder, including higher viscosity and more consistent penetration grades. It met or exceeded the test requirements across the board.
- Replacing 20% of the coarse aggregate with cubical particles and using local ERL bitumen (Mix-3) gave a Marshall stability of 2991 lbs (13.3 kN) and a flow of 8.5 units.
- The volumetric analysis showed that while the conventional mixes stayed within standard tolerances, the imported bitumen with 20% cubical aggregates (Mix-4) produced air voids of only 2.59% and a VFA of 82.65%, which increases the risk of binder bleeding under traffic.
- Local ERL bitumen paired with cubical aggregates gives a mix that is both structurally strong and volumetrically stable, meaning higher load-bearing capacity without compromising durability.
- **Proposed Roadmap for Bangladesh:** A two-level strategy is proposed: (1) *Foundation Level:* stabilize weak subgrade soils with lime and fly ash to raise the CBR; and (2) *Surface Level:* use local ERL bitumen with at least 20% cubical aggregates. Together, these steps would reduce the volume of imported materials needed.

5.2 Recommendations for Practice

From the experimental results, we recommend a two-level approach for road infrastructure in Bangladesh:

1. **Foundation Level (Subgrade Stabilization):** For areas with weak subgrade soils, treating the soil with lime or fly ash is recommended. This raises the California Bearing Ratio (CBR) and allows the pavement layers above to be thinner.
2. **Surface Level (Optimized Bituminous Mix):** The Roads and Highways Department (RHD) should consider adding quality control requirements for imported bitumen and introducing aggregate shape sorting into the specifications. Setting a minimum threshold of 20% cubical coarse aggregates, to be used with local ERL bitumen, would improve the interlocking matrix of the mix.

Making these changes would raise the structural layer coefficient (a_1) of the surfacing course, which means the same structural performance can be achieved with less material. Over time, this reduces lifecycle costs for highway projects.

5.3 Future Work

Several areas remain open for further investigation:

- **Indirect Tensile Strength (ITS) Testing:** Running ITS tests (ASTM D6931) would help evaluate how the different mix combinations resist tensile loading and fracture.

- **Moisture Susceptibility and Climate Resistance:** Retained stability tests (Marshall immersion) or Tensile Strength Ratio (TSR) tests would show how well these mixes resist stripping under conditions similar to Bangladesh's monsoon climate.

- **Rutting and Fatigue Evaluation:** The Hamburg Wheel Tracking Test (AASHTO T 324) could be used to measure rutting depth, and the Four-Point Bending Beam Fatigue Test (AASHTO T 321) would give information on cracking resistance under repeated loading.

- **Evaluation of Polymer Modified Bitumen (PMB):** It would be worth investigating what happens when 5% SBS (Styrene-Butadiene-Styrene) polymer is added to these base bitumens to produce PMB, following LGED specifications. Such a study could show how much polymer modification improves load-bearing capacity and fatigue life.

- **Performance Enhancement of Imported Bitumen:** Since Bangladesh will likely continue importing bitumen, future work should look into ways to improve imported binders. Additives like 5% SBS polymer, crumb rubber, or waste plastics could potentially bring the structural and rheological properties of imported bitumen closer to those of the local product.

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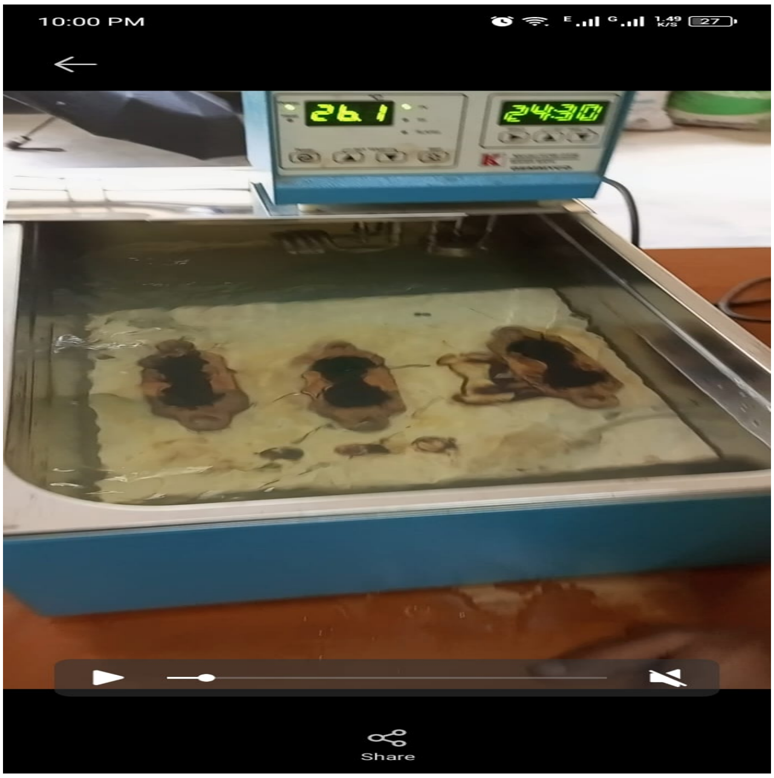
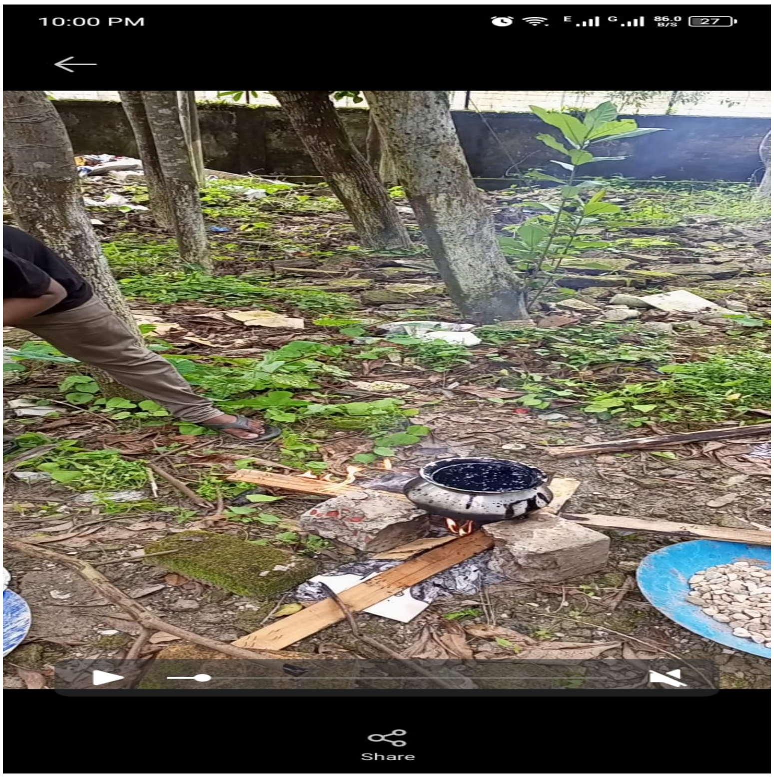
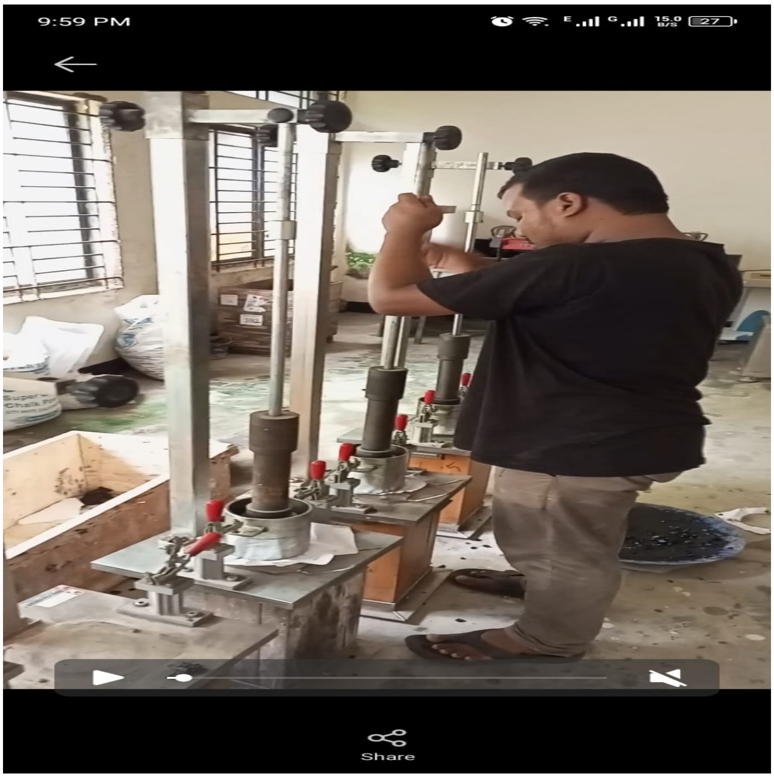
6 Appendix



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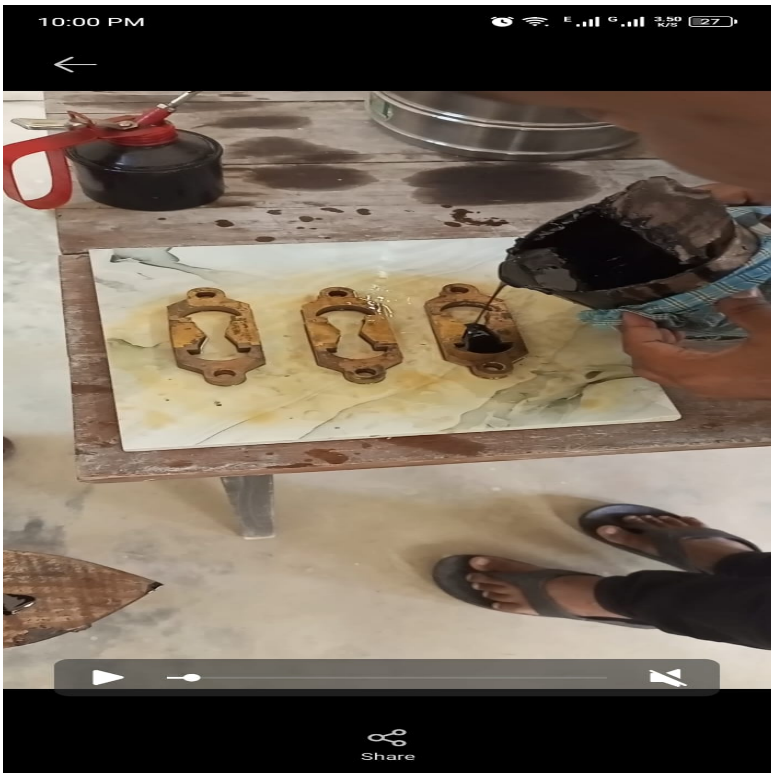
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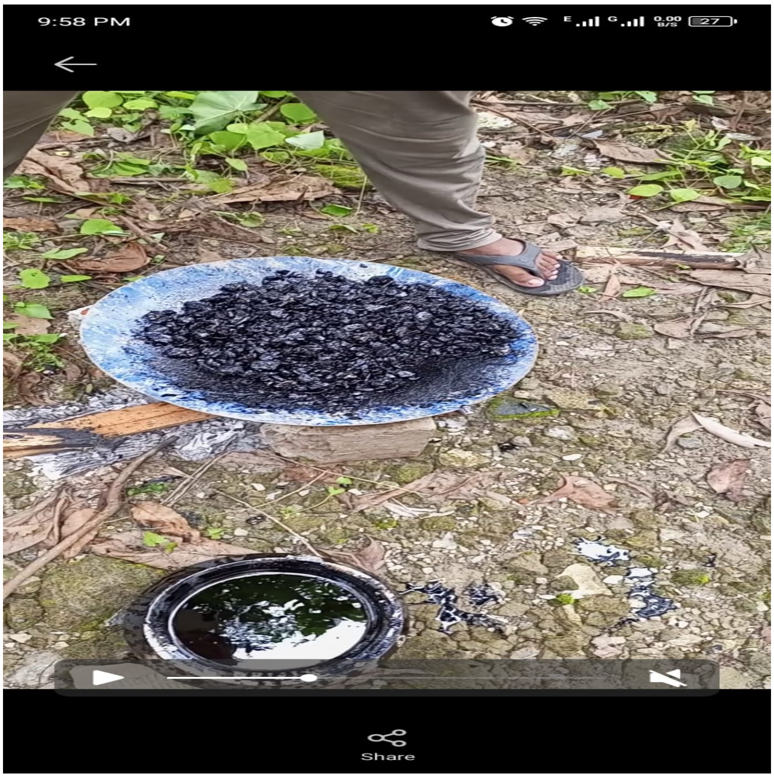
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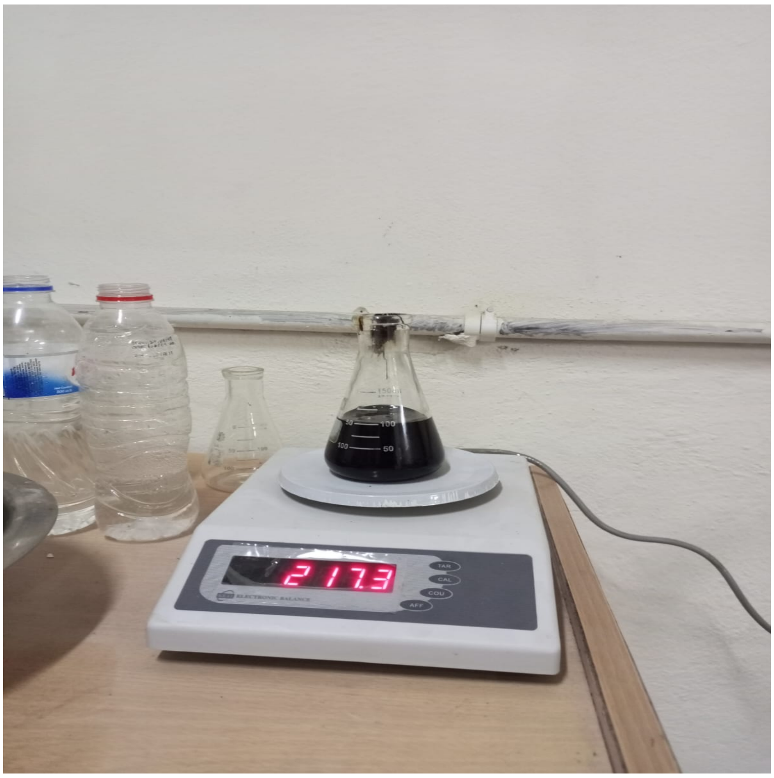
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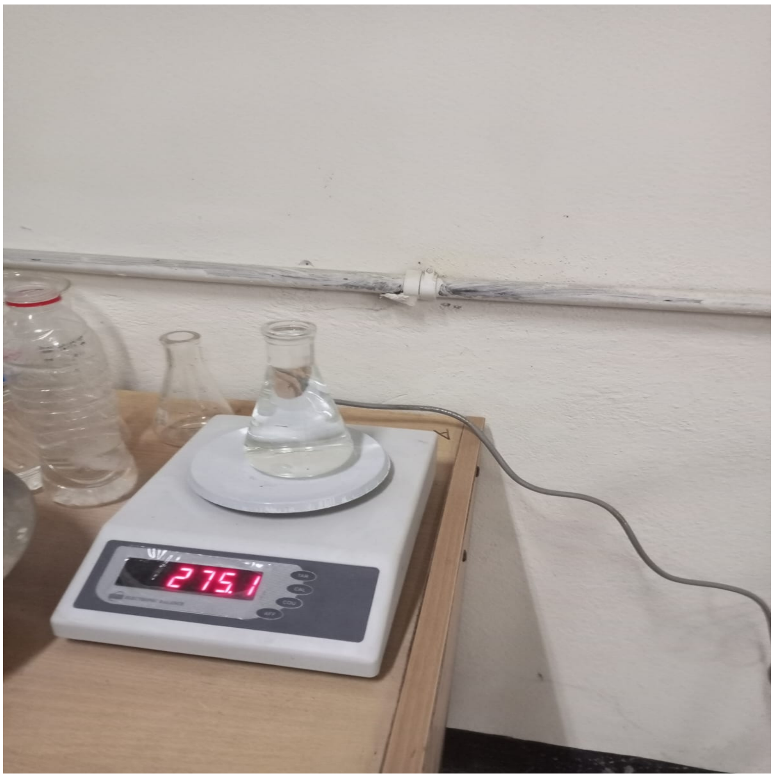
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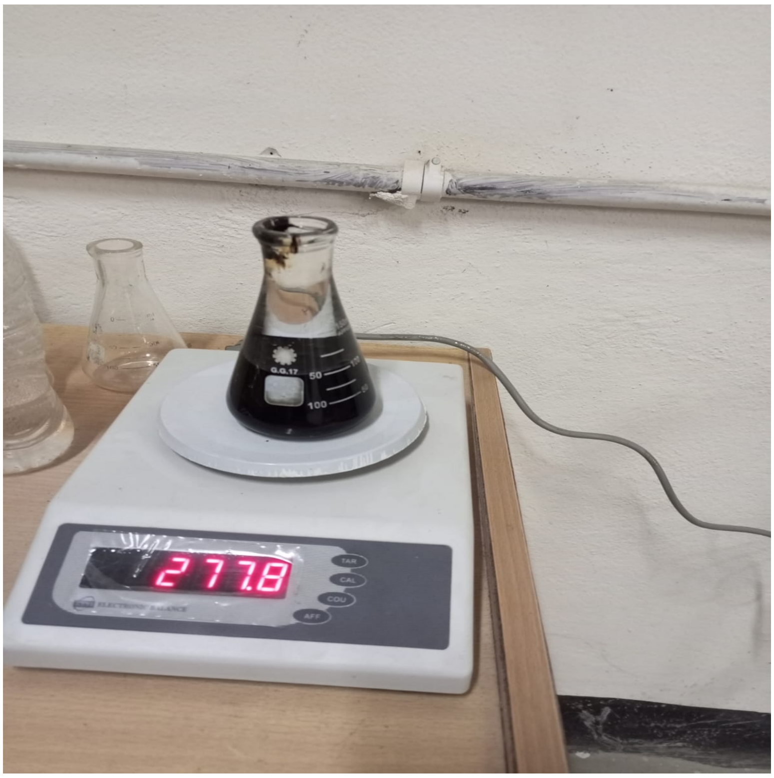
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